MAVEN CAFÉ

ANALYZING 30 DAY PROMOTIONAL OFFERS

Sources

https://www.kaggle.com/datasets/arshmankhalid/caf-rewards-offer-dataset

Goal

The Maven Café sent promotional offers for registered customers in order to enhance sales.

The goal of the analysis is to evaluate the response of the customers to the different offers they received and develop a data-driven strategy for future promotional messaging & targeting.

Data Input

Table Name	Count	Filter	Final Count
Offers	10		
Customers	17,000	2,175 with no data on age, gender and income	14,825
Events	306,534		

Data Input

Offers

There are 10 different offers, grouped into three categories Informational, Buy One Get One (BOGO), Discount.

Customers

14,825 unique existing customers with details on the gender, age, income and tenure as a customer.

Events

306,534 events during the 30-day period. We concluded that every customer could complete several offers and that each offer could have been completed several times by the same costumer.

Data Analysis Process

Data Collection

We gathered the datasets from Kaggle website, downloaded it as CSV files

Data Analysis

We utilized SQL Views to create a robust Abstraction Layer, ensuring data consistency and simplifying the data model for Power BI

Interpretations

Visualizations were built using Power BI and business action points were concluded in order to efficient future campaigns

Data Cleaning

- Some data was stored in a "nested" format (JSON), making it inaccessible for direct analysis.
- Mixed data types and inconsistent formatting prevented accurate calculations.
- We had to utilize JSON_VALUE to parse complex strings into structured columns and then we converted raw text into calculated fields .

```
create view events_clean as
select * ,json_value (replace(replace(value,char(39) ,'"' ),'offer id', 'offer_id'),'$.offer_id') as offerId,
cast(json_value (replace(value,char(39) ,'"' ),'$.amount') as float) as trans_amount,
cast(json_value (replace(value,char(39) ,'"' ),'$.reward') as float) as reward
from events
```

- In order to analyze data regarding customers' age, income and tenure we had to create bins in order to examine the data properly and produce accurate conclusions.
- We utilized SQL Views to create a robust Abstraction Layer, ensuring data consistency and simplifying the data model for Power BI while maintaining a 'Single Source of Truth' for all business logic

```sql
create view customers_clean as
with cte_age_bins as (
    select *, floor(age/10) * 10 as age_index
,CAST(FLOOR(age/10) * 10 AS VARCHAR) + '-' + CAST(FLOOR(age/10) * 10 + 9 AS VARCHAR) AS age_bin
    from customers
)

, cte_income as
    (select *
        , floor(income/10000) * 10000 as income_index
        , cast(floor(income/10000) * 10000 as varchar) + '-' + cast(floor(income/10000) * 10000 + 9000 as varchar) as income_bin

    from customers
    )

, cte_membership as
    (SELECT *,

    FLOOR(DATEDIFF(day, c.became_member_on, '2018-07-26') / 365.0) AS membership_index,
    'Year ' + CAST(FLOOR(DATEDIFF(day, c.became_member_on, '2018-07-26') / 365.0) + 1 AS VARCHAR) AS membership_bin
FROM customers c
    )

SELECT a.*, income_index, b.income_bin, c.membership_index, c.membership_bin
FROM cte_age_bins a
join cte_income b on a.customer_id = b.customer_id
join cte_membership c on a.customer_id = c.customer_id
where a.age != 118
```

Data Cleaning

Business Questions

- **Target Audience Analysis: Optimization of Future Promotions**

  Which customer segment demonstrated the strongest response to the campaign, and based on their characteristics, what is the most effective offer structure (Reward, Difficulty, and Duration) for future targeting?

- **Re-Marketing Feasibility: Identifying High-Revenue Drivers for Future Campaigns**

  Based on current performance metrics, is it recommended to re-market these offers in the future? Specifically, which campaign generated the highest total revenue to justify its recurrence?

- **Conversion Efficiency Analysis: Identifying High-Completion Incentives**

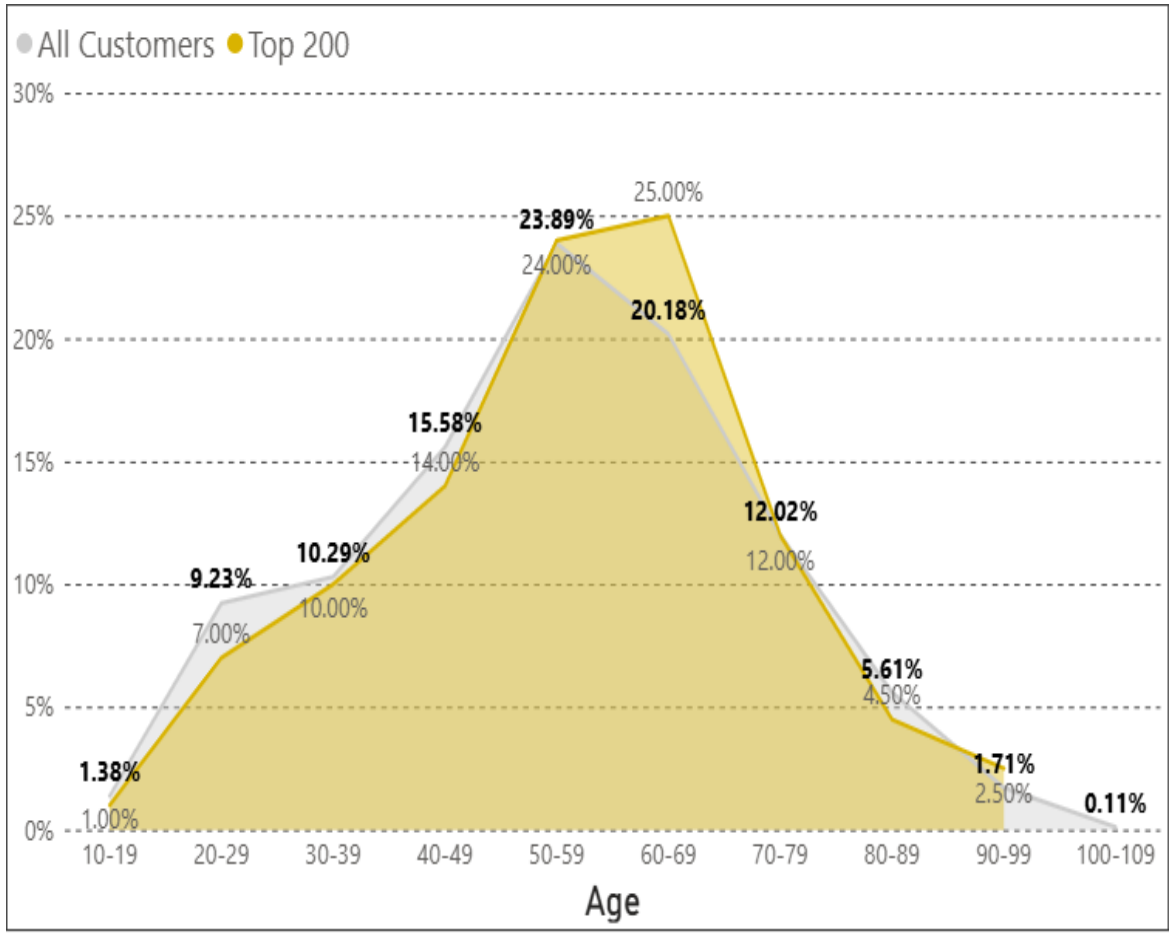
  Which specific offer mechanics and delivery timings achieved the highest conversion rates from "Offer Received" to "Offer Completed," and what insights can be derived from the peak point within the conversion funnel?

Q1

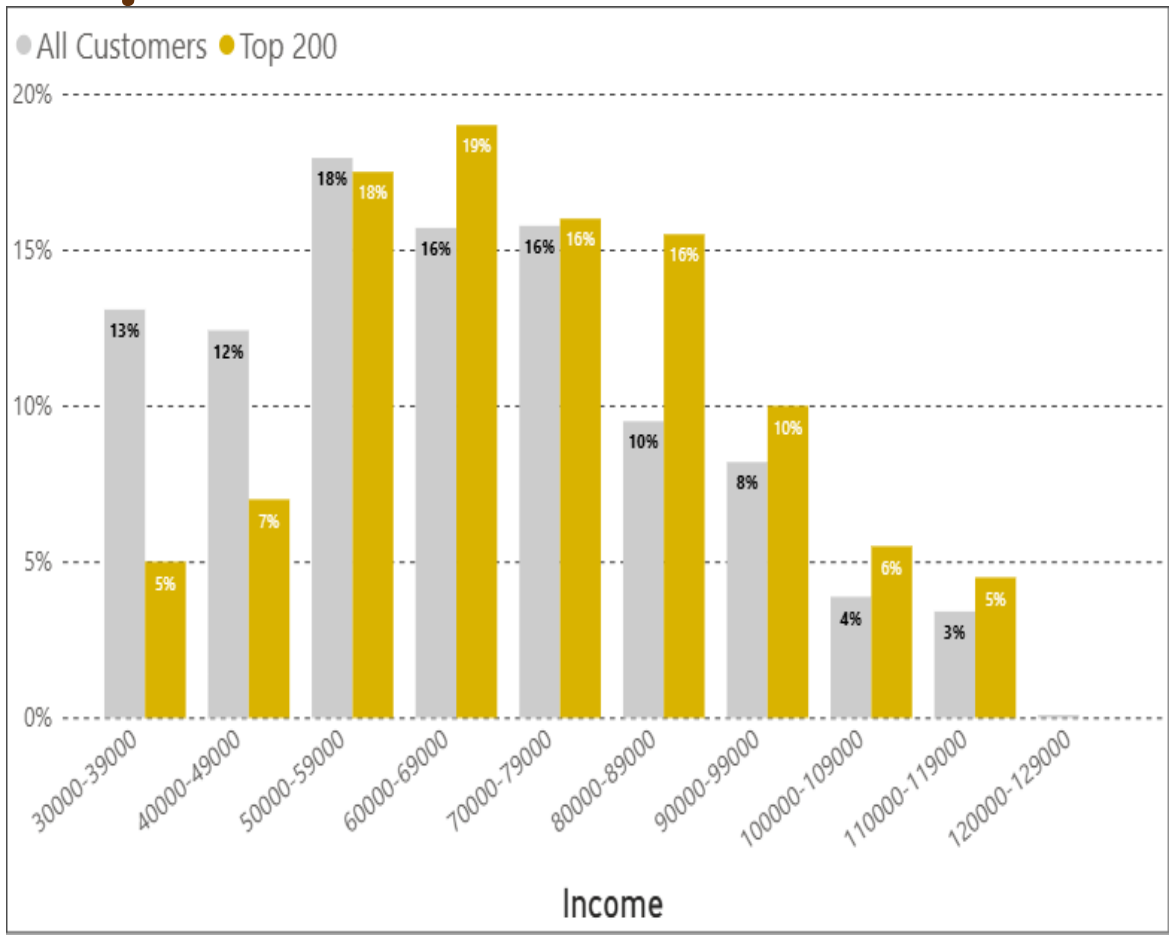
Which customer segment demonstrated the strongest response to the campaign and what is the most effective offer structure for future targeting?

Top 200 Customers - Age Distribution

- No significant variance observed in age distribution between the top 200 group and the rest of the population.
- The average age is 56, 49% of the customers are within the 50-69 age range.

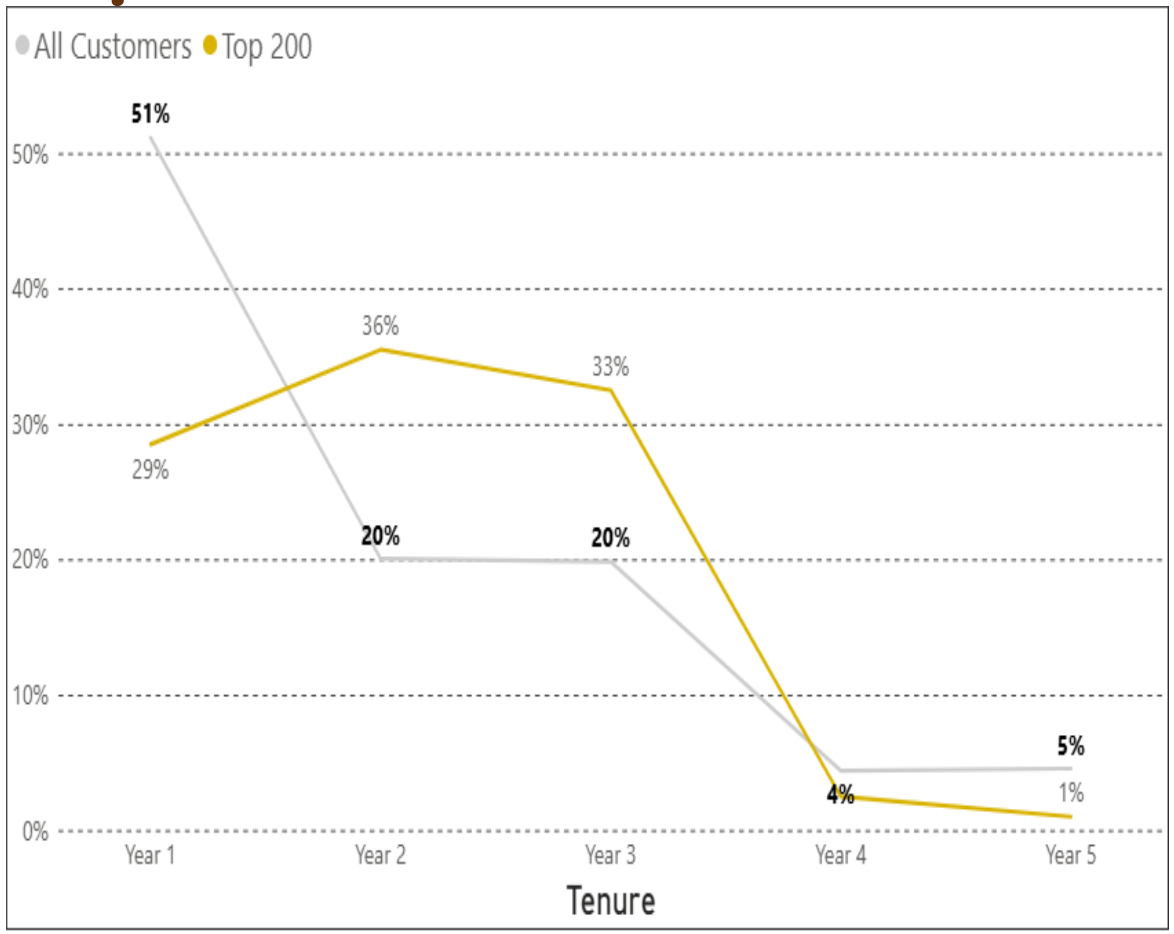


Top 200 Customers - Income Distribution



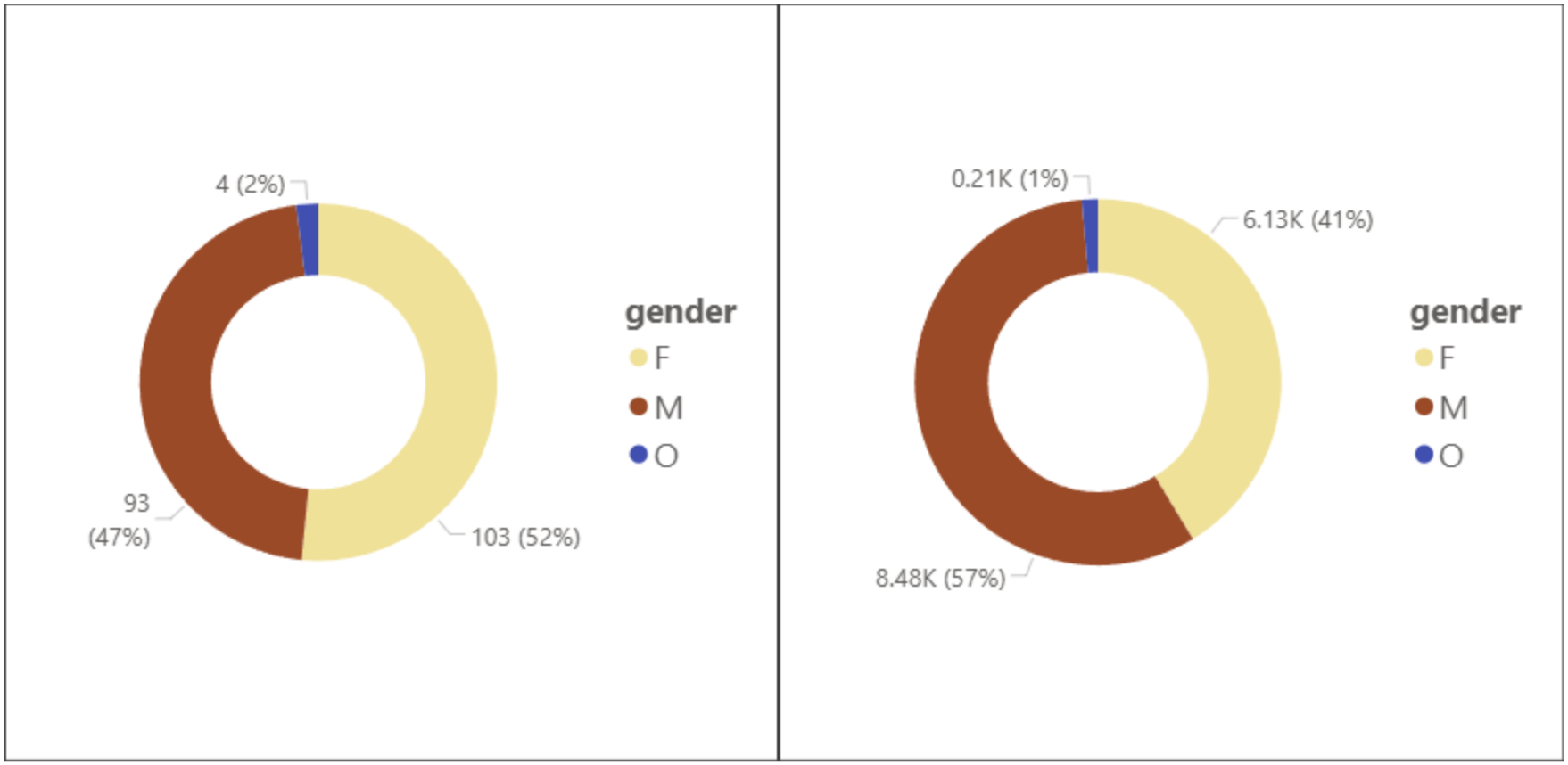
- The average income is 71,760$, 68% are within the 50K$-89K$ income range.
- In the two lowest income brackets, the Top 200 group show lower representation compared to the general customer base.
- There is a 6% over-representation to the top 200 group in the 80K$-89K$ brackets compared to the total customer base.

Top 200 Customers – Tenure Distribution



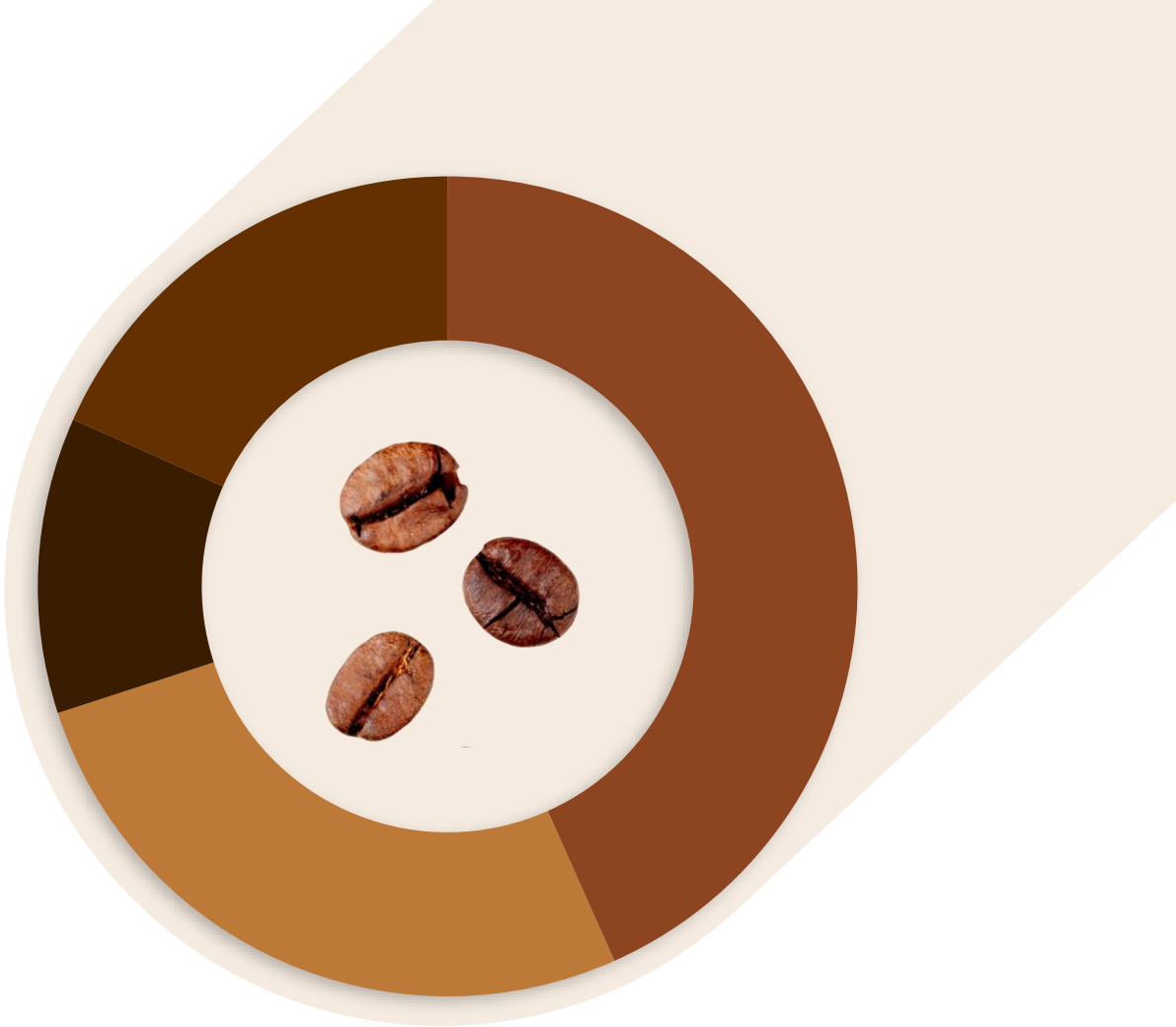
- 91% are with a tenure lower than 4 years.
- In the 2-3 years brackets, the Top 200 group show higher representation compared to the general customer base.
- In the first year bracket, the top 200 group show lower representation compared to the total customer base.

Top 200 Customers - Gender Distribution



- In the Top 200 group, female customers (F) represent 52% of the population, an 11% increase from the 41% observed in the general customer base.
- Conversely, The male segment accounts for 47% of the Top 200, representing a 10% decrease compared to their concentration in the total customer base.

Top 200 Customers – Summary



Core audience (49%) consistently falling between the ages of 50 and 69.

Top performing customers are primarily concentrated in the $50K–$89K range (68%).

The vast majority of top customers (91%) have a tenure of less than 4 years.

Top 200 Customers – Strategic Recommendations & Action Points

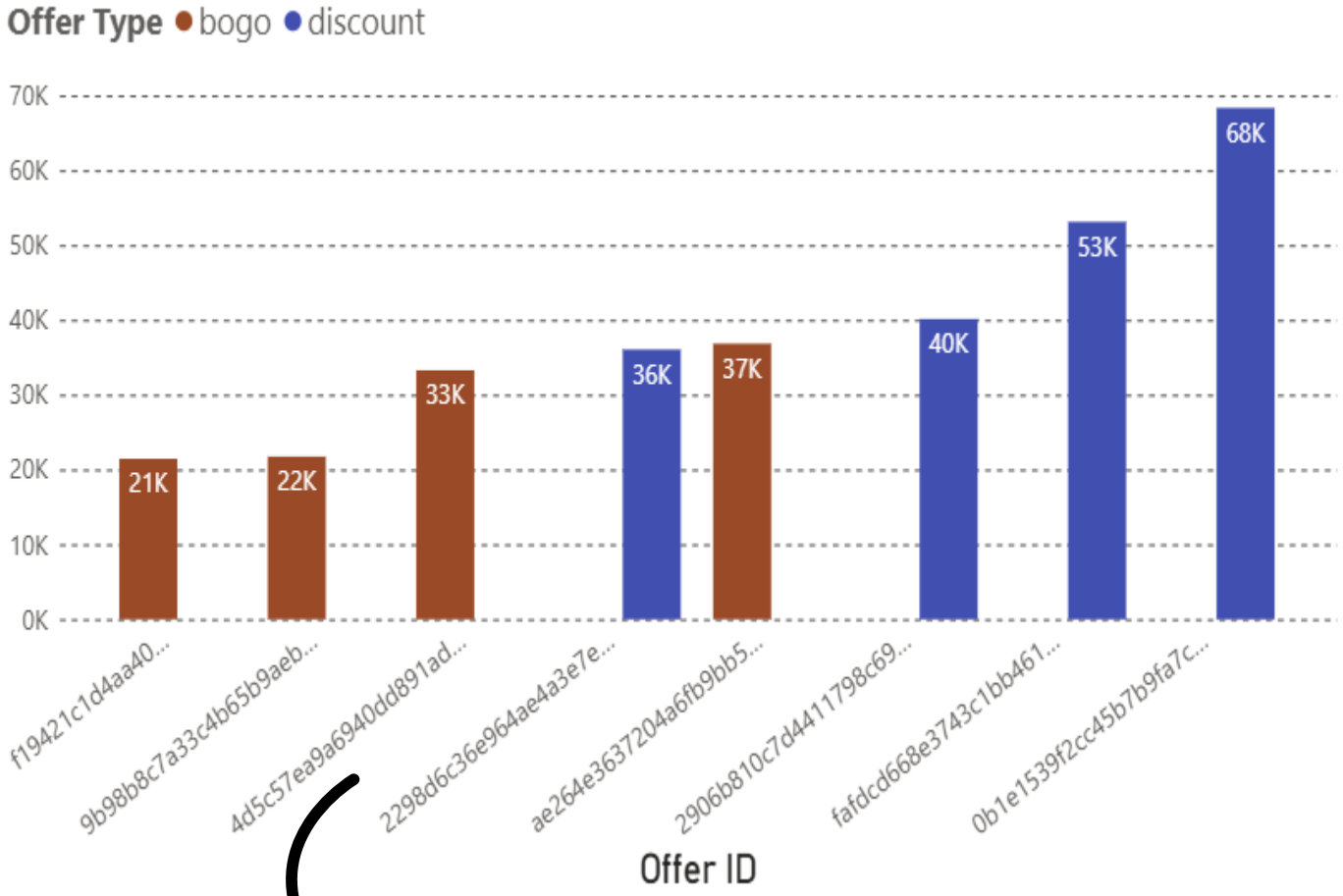
- Maintain the current offer strategy for the 50–69 age group.
- Prioritize new prospects falling within the $60K–$89K income brackets.
- Increase loyalty initiatives and personalized engagement toward the end of Year 1.
- Develop tailored, alternative offers, specifically targeting:
  - Customers with a tenure exceeding 4 years.
  - Individuals with an annual income below $49K.
  - Demographics under age 50 or above age 69.

Q2

Which campaign generated the highest total revenue to justify its recurrence?

Completed Offers - Offer Types And Revenue

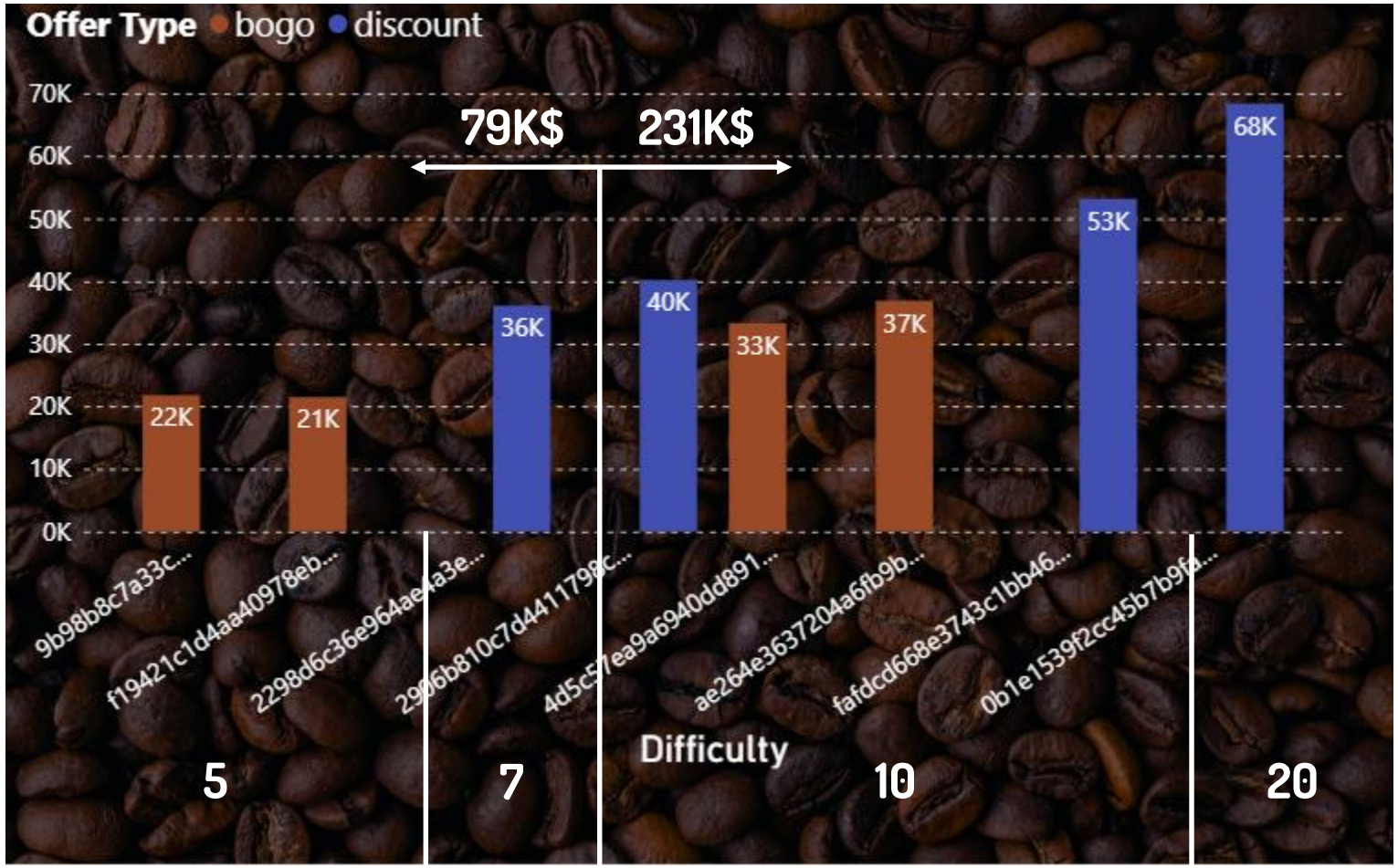
BOGO offers provided the remaining 37% of the revenue share, contributing $113,000 to the overall total



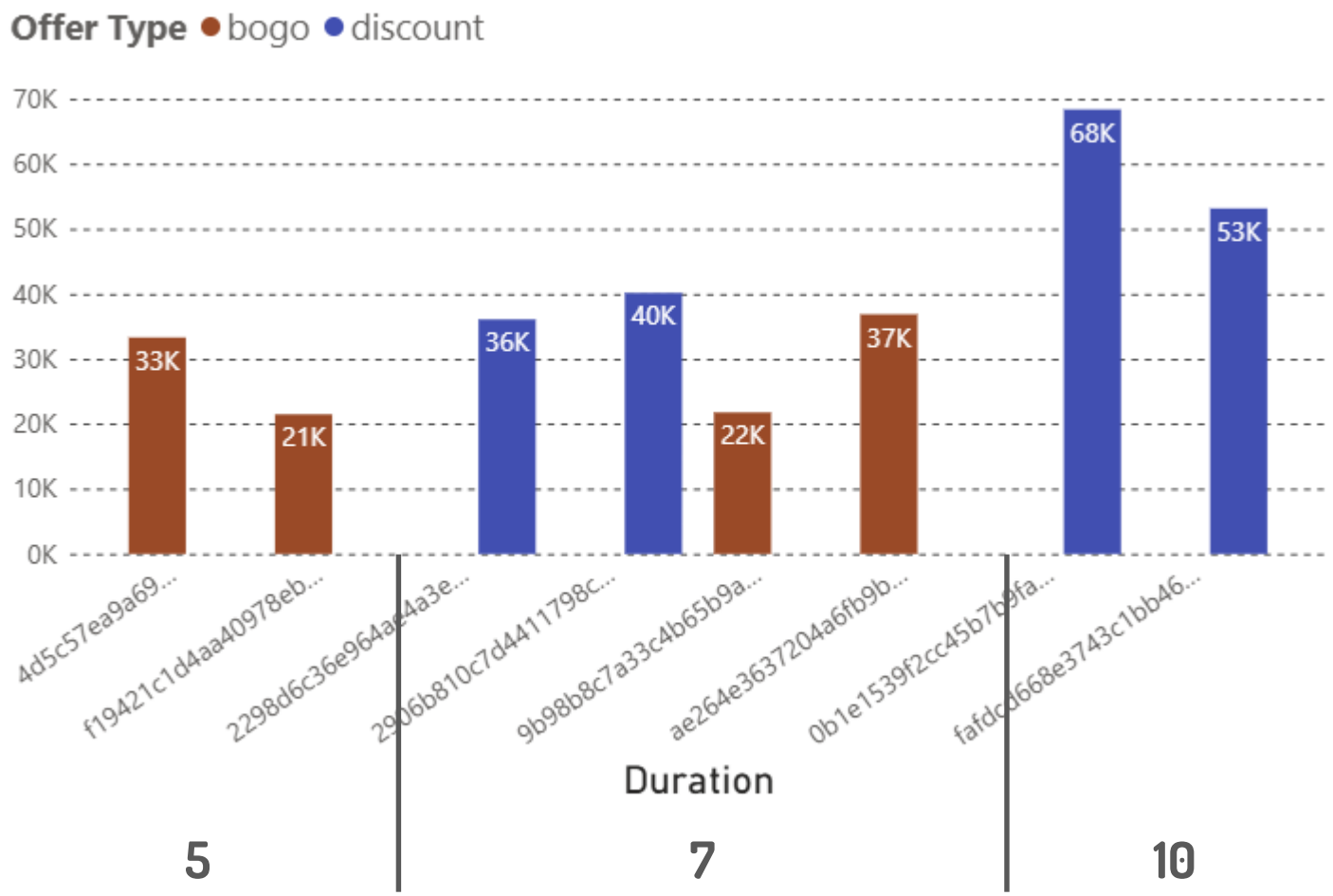
Discount-based incentives emerged as the dominant revenue source, generating $197,000—which accounts for 63% of the total revenue earned from completed offers.

None of the informational offers resulted in a completion, contributing zero revenue to this segment.

Completed Offers - Difficulty And Revenue



- Offers 10 and 20 difficulty levels generated 74% of the total revenue.
- Offers with 5 and 7 difficulty levels representing the remaining 26% of the revenue.

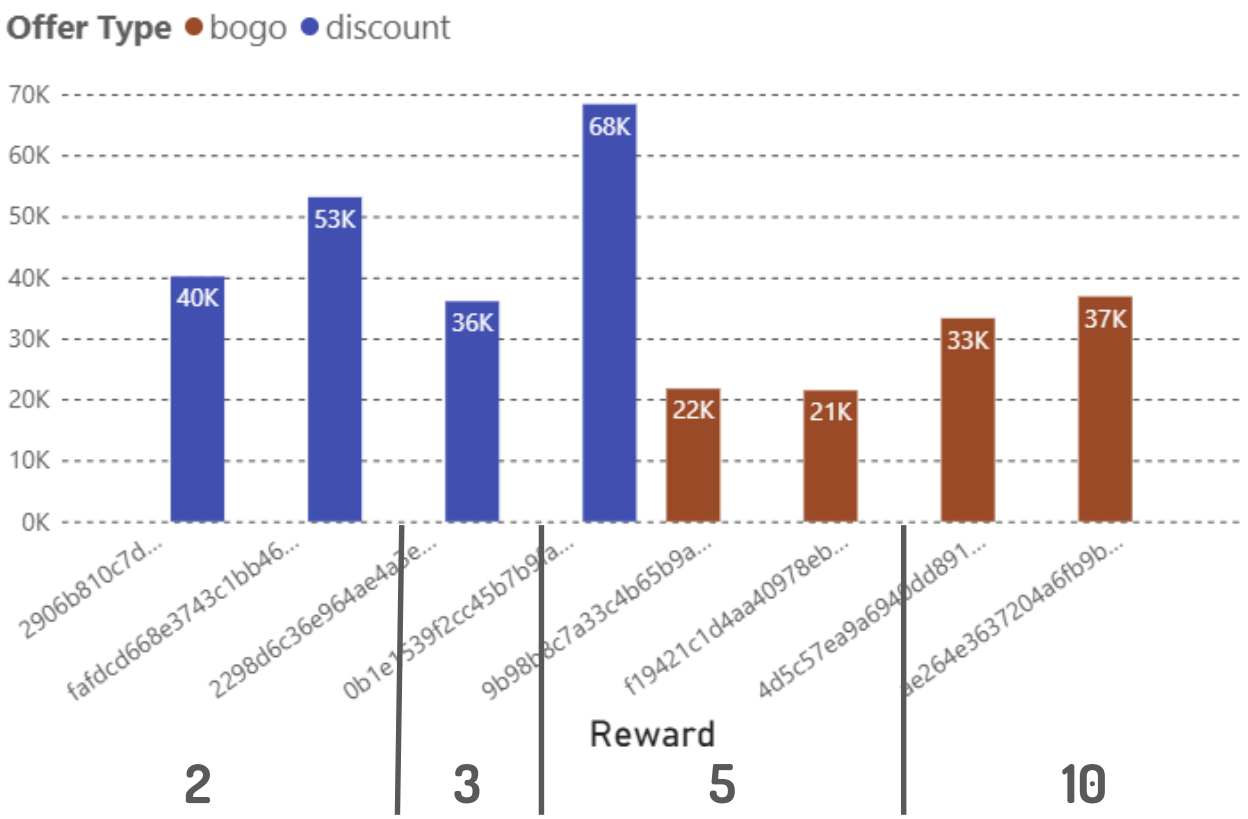


Completed Offers – Duration And Revenue

- Offers with a 10-day duration generated $121,000, accounting for 39% of the total revenue from completed offers. Within that category revenue was generated exclusively by discount offers.
- Offers with a 7-day window contributed $135,000 (43% of total revenue), with 56% of it originating from discount incentives and the remainder from BOGO ones.

Completed Offers – Reward And Revenue

- Reward values of $2 and $3 emerged as the most efficient revenue drivers, generating $129,000 (41% of the total revenue) exclusively through discount-based offers.
- The $5 reward tier contributed $111,000 (35% of the total revenue), with 61% of that amount came from the discount offers and the rest from BOGO offers.



Completed Offers - Summary

Discount offers are significantly more effective at driving high-value conversions.

There is a strong customer willingness to engage with more challenging offers, provided the incentive is sufficient.

The $5 reward category is a major revenue contributor..

There is a clear correlation between longer duration (10 days) and discount offer success.

Completed Offers – Strategic Recommendations & Action Points

01 Informational offers should be re-evaluated as they generated no revenue.

02 Marketing efforts should prioritize discount-based promotions.

03 Future campaigns should continue to utilize offers with difficulty ratings of 10 and 20.

04 Prioritizing a 10-day duration for discount-based offers.

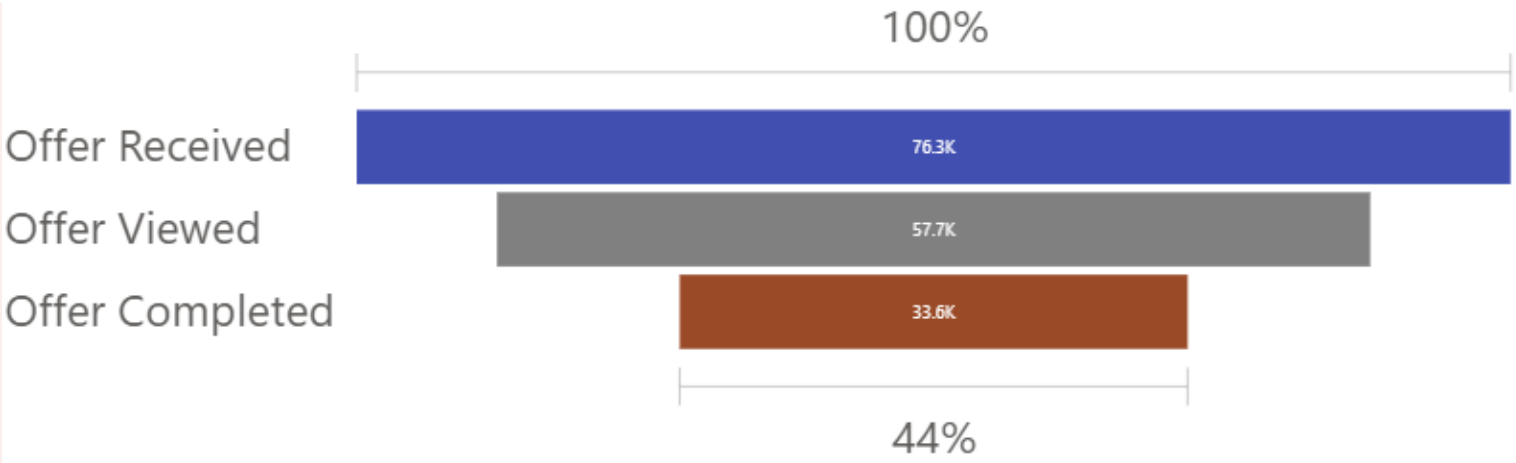
05 For 7-day duration, continue utilizing a hybrid approach of discount and BOGO.

06 Prioritizing discount-based offers with the 5$ reward incentives

Q3

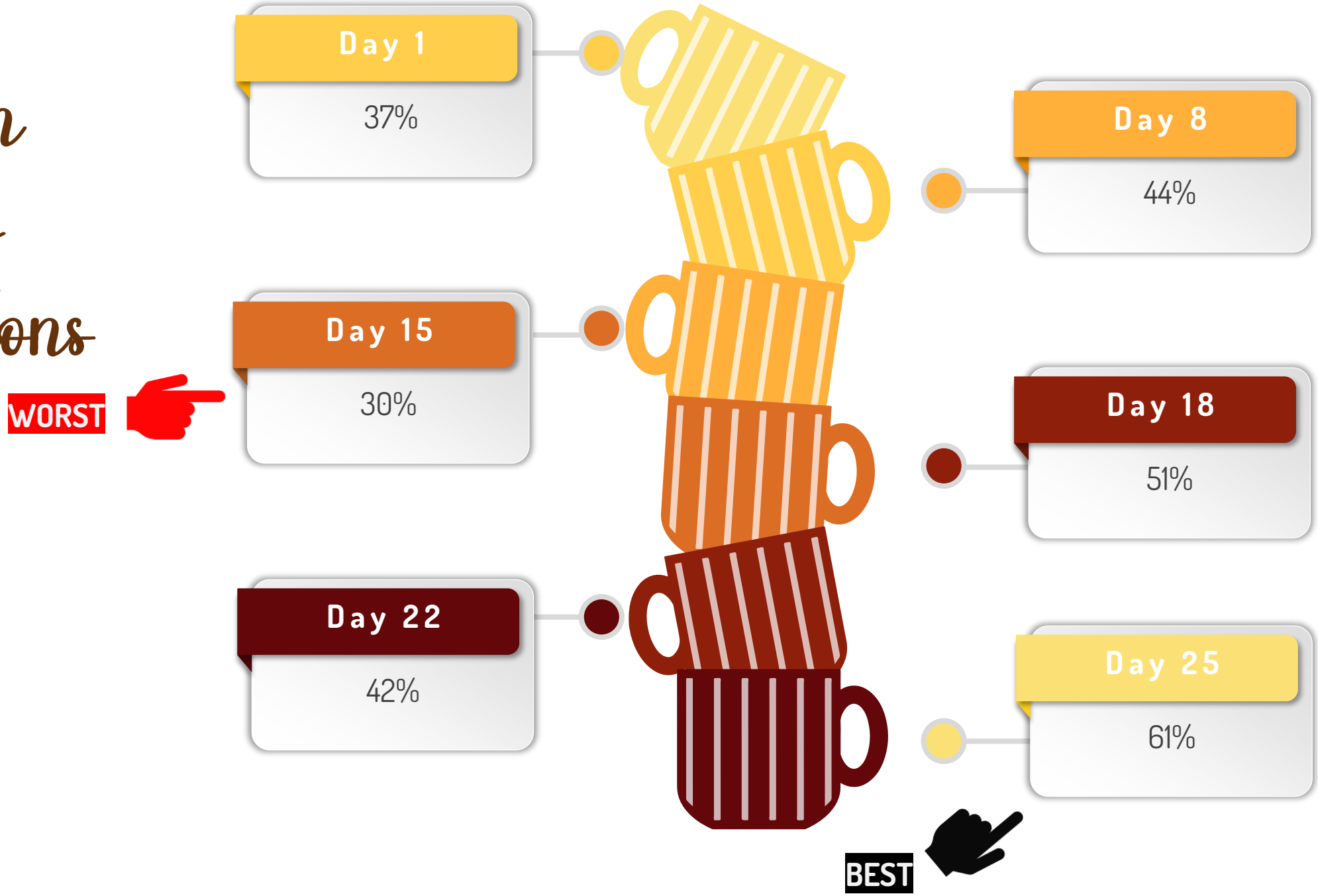
Which specific offer mechanics and delivery timings achieved the highest
conversion rates from "Offer Received" to "Offer Completed"?

Total Offers Conversion Rate



- Out of 76,277 total offers received, 75% were viewed by customers. Of those viewed, 58% reached successful completion.
- The total completion rate relative to the total number of offers sent is 44%.

Offers Conversion Rate By Distributions



Final Recommendations

Focus marketing spend on low-reward Discount offers

Scale Discounts

Prioritize loyalty programs for customers in their 2nd/3rd year

Retention Focus

Replicate the peak period's timing and mix for 61% conversion

Adopt the "Time 25" Model

Strategic Summary

Target Audience Insights

- **The High-Value Segment:** 52% Female, primarily aged 50-69, earning $50K–$89K.
- **Peak Maturity:** Customer value is highest in Years 2 and 3.

Winning Offer Strategy

- **Discount Dominance:** Generated 63% of revenue.
- **Maximum ROI:** $2–$3 rewards and 10-day durations performed best.
- **Engagement:** High-difficulty offers drove 74% of total profit.

Conversion Funnel Optimization

- **Overall Completion:** 44% successful conversion rate.
- **Record Growth:** Conversion efficiency surged from 30% to a 61% peak.

THANK
YOU!